

```
:- [lab5lmc3].
```

```
/* Notez suma ordinala cu + si ridicarea la putere cu ^. Sa construim cateva poseturi  
(P,OrdP), introducand multimea suport P si relatia de succesiune, apoi obtinand din acestea  
relatia de ordine OrdP cu predicatul orddinsucc: */
```

```
% A = L2^2+L2: suma ordinala dintre romb si lantul cu doua elemente:
```

```
posetA(A,OrdA) :- A=[0,a,b,c,1],  
                  orddinsucc([(0,a),(0,b),(a,c),(b,c),(c,1)],A,OrdA).
```

```
% B = "V rasturnat":
```

```
posetB(B,OrdB) :- B=[u,v,1], orddinsucc([(u,1),(v,1)],B,OrdB).
```

```
% L2^2: rombul:
```

```
romb(R,OrdR) :- R=[0,a,b,1], orddinsucc([(0,a),(0,b),(a,1),(b,1)],R,OrdR).
```

```
% L2: lantul cu doua elemente:
```

```
l2([0,1],OrdL2) :- orddinsucc([(0,1)],[0,1],OrdL2).
```

```
% L2+L2^2+L2:
```

```
latL(L,OrdL) :- L=[0,a,x,y,b,1],  
                 orddinsucc([(0,a),(a,x),(a,y),(x,b),(y,b),(b,1)],L,OrdL).
```

```
/* Predicat care testeaza daca o functie  $F:A \rightarrow B$  pastreaza relatii binare  $R \subseteq A^2$  si  $S \subseteq B^2$ ,  
mai precis daca duce pe  $R$  in  $S$ , adica are proprietatea ca, pentru orice  $x, y$  in  $A$ , daca  $(x, y)$   
in  $R$ , atunci  $(F(x), F(y))$  in  $S$ : */
```

```
pastrrel(F,R,S) :- not((member((X,Y),R), member((X,FX),F), member((Y,FY),F),  
                        not(member((FX,FY),S)))).
```

```
% Functiile crescatoare  $F:P \rightarrow Q$  intre doua poseturi  $(P, \text{OrdP})$  si  $(Q, \text{OrdQ})$ :
```

```
fctcresc(F,P,OrdP,Q,OrdQ) :- functie(F,P,Q), pastrrel(F,OrdP,OrdQ).
```

```
% Functiile strict crescatoare  $F:P \rightarrow Q$  intre doua poseturi  $(P, \text{OrdP})$  si  $(Q, \text{OrdQ})$ :
```

```
fctstrcresc(F,P,OrdP,Q,OrdQ) :- functie(F,P,Q), ordstrdinord(OrdStrP,OrdP),  
ordstrdinord(OrdStrQ,OrdQ), pastrrel(F,OrdStrP,OrdStrQ).
```

```
/* Interogati:
```

```
?- posetB(B,OrdB), l2(L2,OrdL2), fctcresc(F,B,OrdB,L2,OrdL2).
```

```
?- posetB(B,OrdB), l2(L2,OrdL2), fctstrcresc(F,B,OrdB,L2,OrdL2).
```

```
?- posetA(A,OrdA), posetB(B,OrdB), fctstrcresc(F,A,OrdA,B,OrdB).
```

```
si dati ;/Next pentru a obtine toate solutiile.
```

```
*/
```

```
% Multimea functiilor crescatoare injective de la posetul A la posetul B:
```

```
fctlecrecscinjAlaB(LF) :- posetA(A,OrdA), posetB(B,OrdB),
```

```
        setof(F, (fctcresc(F,A,OrdA,B,OrdB), injectiv(F)), LF), !.  
fctlecrescinjAlaB([]).
```

```
% Multimea functiilor crescatoare surjective de la posetul A la posetul B:
```

```
fctlecrescsurjAlaB(LF) :- posetA(A,OrdA), posetB(B,OrdB),  
        setof(F, (fctcresc(F,A,OrdA,B,OrdB), surjectiv(F,B)), LF), !.  
fctlecrescsurjAlaB([]).
```

```
/* Interogati:  
?- fctlecrescinjAlaB(LF).  
?- fctlecrescsurjAlaB(LF).  
*/
```

```
/* Determinarea minorantilor, respectiv a majorantilor M ai unei submultimi S a unui poset  
(P,OrdP): */
```

```
minoreaza(M,S,Ord) :- not((member(X,S), not(member((M,X),Ord)))).
```

```
minorant(M,S,P,OrdP) :- member(M,P), minoreaza(M,S,OrdP).
```

```
minorantii(S,P,OrdP,LM) :- setof(M, minorant(M,S,P,OrdP), LM), !.  
minorantii(_,_,_,[]).
```

```
majoreaza(M,S,Ord) :- not((member(X,S), not(member((X,M),Ord)))).
```

```
majorant(M,S,P,OrdP) :- member(M,P), majoreaza(M,S,OrdP).
```

```
majorantii(S,P,OrdP,LM) :- setof(M, majorant(M,S,P,OrdP), LM), !.  
majorantii(_,_,_,[]).
```

```
/* Determinarea minimului, respectiv a maximului M unei multimi S raportat la ordinea Ord;  
ca si in cazul predicatelor minoreaza si majoreaza, Ord poate fi o ordine pe S sau pe o  
multime care include pe S; de fapt, pentru ca aceste predicate sa functioneze, este  
suficient ca Ord sa fie o lista care include o relatie de ordine pe S: */
```

```
min(S,Ord,M) :- minorant(M,S,S,Ord).
```

```
max(S,Ord,M) :- majorant(M,S,S,Ord).
```

```
/* Determinarea infimumului, respectiv a supremumului M al unei submultimi S a unui poset  
(P,OrdP): */
```

```
inf(S,P,OrdP,M) :- minorantii(S,P,OrdP,LM), max(LM,OrdP,M).
```

```
sup(S,P,OrdP,M) :- majorantii(S,P,OrdP,LM), min(LM,OrdP,M).
```

```
/* Interogati:
```

```
?- posetA(A,OrdA), minorantii([a,b],A,OrdA,Minorantii), majorantii([a,b],A,OrdA,Majorantii).
```

```
?- posetA(A,OrdA), inf([a,b],A,OrdA,Inf), sup([a,b],A,OrdA,Sup).
```

```
?- posetA(A,OrdA), min([a,b],OrdA,Min).
```

```
?- posetA(A,OrdA), max([a,b],OrdA,Max).
```

```
?- posetA(A,OrdA), min([0,a,b,c],OrdA,Min), max([0,a,b,c],OrdA,Max).
```

```
?- posetA(A,OrdA), min(A,OrdA,Min), max(A,OrdA,Max).
```

```

?- posetB(B,OrdB), min(B,OrdB,Min).
?- posetB(B,OrdB), max(B,OrdB,Max).
*/

% Multimea functiilor crescatoare de la  $L^2$  la  $L$ :

fctlecrescRomblaL2(LF) :- romb(R,OrdR), l2(L2,OrdL2),
                        setof(F, fctcresc(F,R,OrdR,L2,OrdL2), LF), !.
fctlecrescRomblaL2([]).

/* Interogati:
?- fctlecrescRomblaL2(LF), afislista(LF), length(LF,NrFct).
*/

/* Sa determinam daca un poset (L,OrdL) este latice (Ore), respectiv latice marginita,
respectiv latice marginita complementata: */

latice(L,OrdL) :- not((member(X,L), member(Y,L),
                      not((inf([X,Y],L,OrdL,_), sup([X,Y],L,OrdL,_))))).

latmarg(L,OrdL) :- latice(L,OrdL), min(L,OrdL,_), max(L,OrdL,_).

latmargcomplem(L,OrdL) :- latice(L,OrdL), min(L,OrdL,Zero), max(L,OrdL,Unu),
                          not((member(X,L), not((member(Y,L),
                          inf([X,Y],L,OrdL,Zero), sup([X,Y],L,OrdL,Unu))))).

/* Interogati:

```

```
?- latL(L,OrdL), latice(L,OrdL).
?- latL(L,OrdL), latmarg(L,OrdL).
?- latL(L,OrdL), latmargcomplem(L,OrdL).
?- posetA(A,OrdA), latmarg(A,OrdA).
?- posetA(A,OrdA), latmargcomplem(A,OrdA).
?- posetB(B,OrdB), latice(B,OrdB).
?- romb(R,OrdR), latmargcomplem(R,OrdR).
*/
```

```
/* Determinarea sublaticilor, respectiv a sublaticilor marginite S, ale unei latici,
respectiv latici marginite date prin laticea sa (Ore) subiacenta (L,OrdL): */
```

```
sublat(S,L,OrdL) :- sublista(S,L), not((member(X,S), member(Y,S),
    inf([X,Y],L,OrdL,XsiY), sup([X,Y],L,OrdL,XsauY),
    not((member(XsiY,S), member(XsauY,S)))))).
```

```
sublatmarg(S,L,OrdL) :- sublat(S,L,OrdL),
    min(L,OrdL,Zero), max(L,OrdL,Unu), member(Zero,S), member(Unu,S).
```

```
% Sublaticile si sublaticile marginite ale lui L2+L2^2+L2:
```

```
sublaticileL(LS) :- latL(L,OrdL), setof(S, sublat(S,L,OrdL), LS), !.
sublaticileL([]).
```

```
sublaticilemargL(LS) :- latL(L,OrdL), setof(S, sublatmarg(S,L,OrdL), LS), !.
sublaticilemargL([]).
```



```
/* Interogati:  
?- sublaticileL(LS), afislista(LS), length(LS,NrSublat).  
?- sublaticilemargL(LS), afislista(LS), length(LS,NrSublatmarg).  
*/
```